



PRAYAS

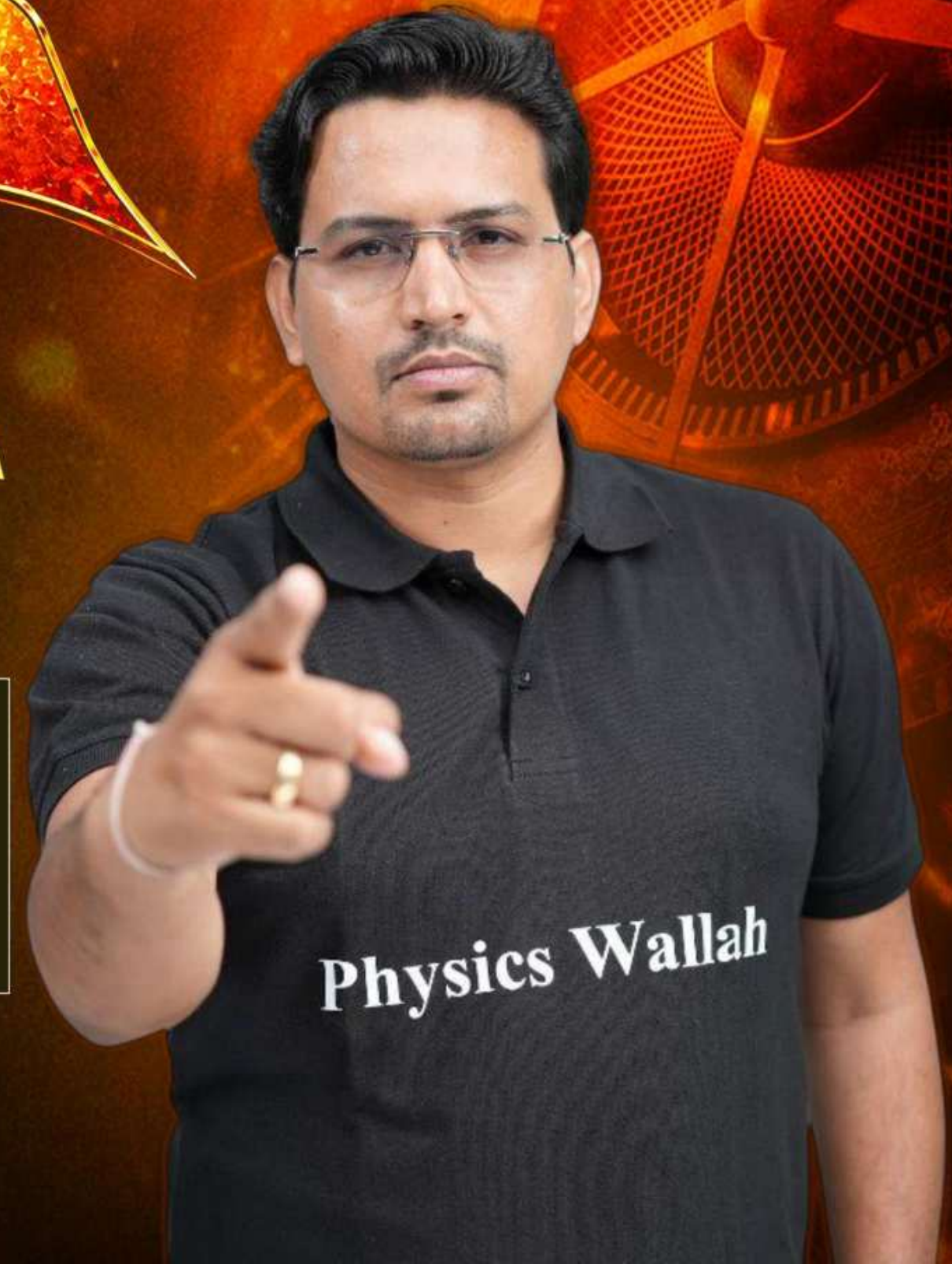
2.0 2027

Physics

Lecture No.- 05

Motion in a Straight Line

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Topics to be covered



- 1 Numerical Solving on Accelerated Motion.
- 2 Miscellaneous Problems.
- 3 Displacement of Particle in n^{th} second.



Rahul Yadav Sir Physics

WhatsApp channel



70% ↑

30% ↓

DPP NO. (4) 60

✓ (A) Above 50 (40%)

✓ (B) 40-50 (24%)

✓ (C) 30-40 (16%)

(d) 20-30 (7%)

(e) Below 20 (12%)



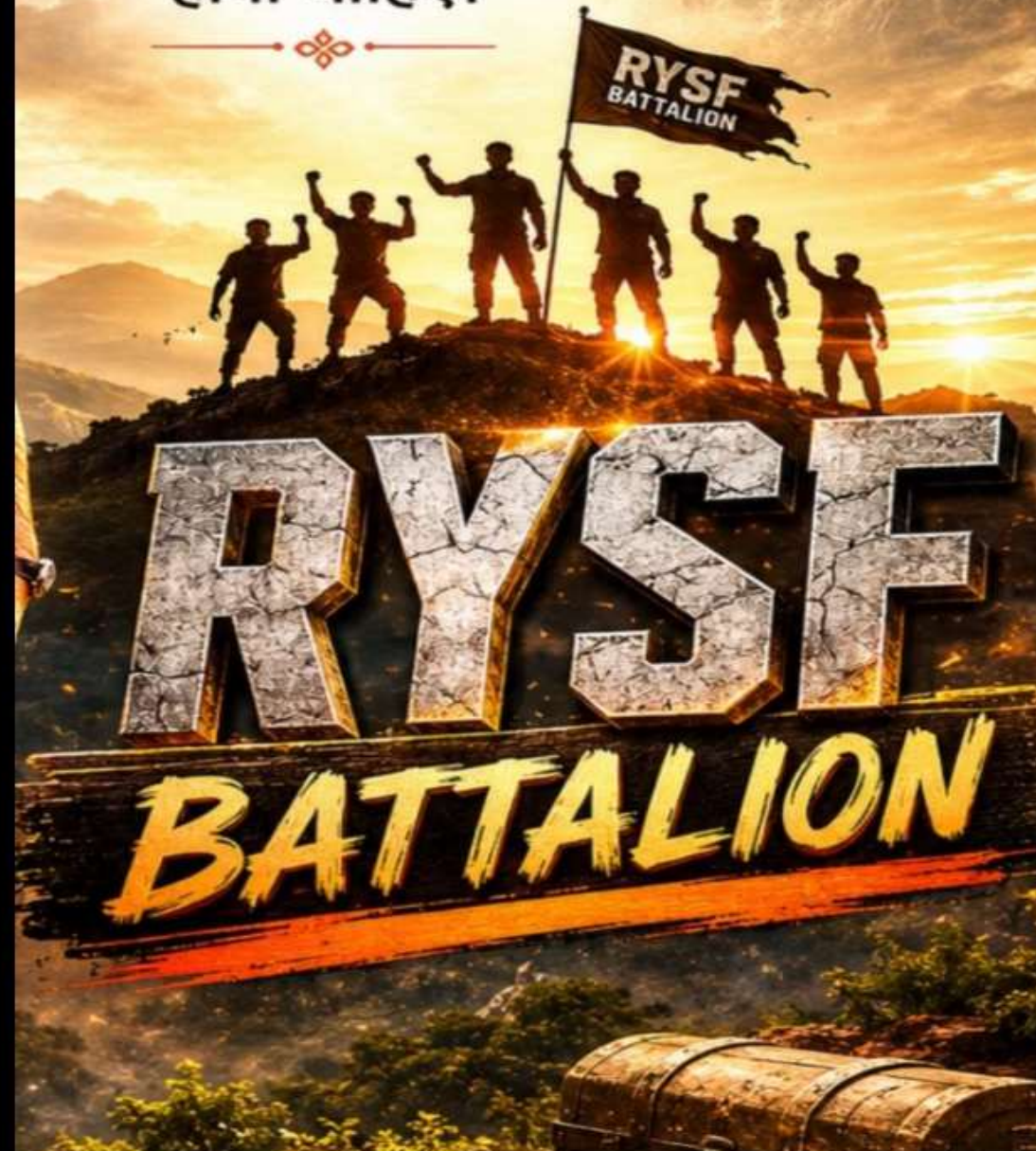
Recap of previous lecture



- 1 Homework Discussion
- 2 Acceleration
- 3 Conditions of Speeding Up and Speeding Down
- 4 Uniform Accelerated Motion.
- 5 Numerical Solving.

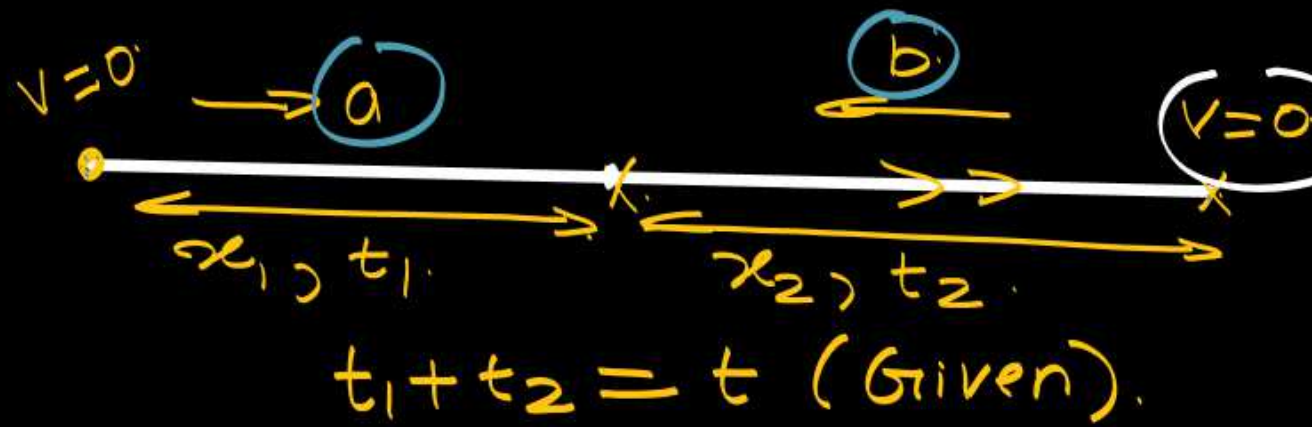
“

कठिनाईयाँ आएंगी,
मन भी भटकेगा,
लेकिन तुम्हारा **लक्ष्य**
तुम्हारी प्राथमिकता
होनी चाहिए।



Question

Homework → 201.



A car accelerates from rest at a constant rate a for some time after which it decelerates at a constant rate b to come to rest. If the total time elapsed is t seconds, the total distance travelled is:

$$s = x_1 + x_2 = ?$$

$$2(a) \quad x_1 = \cancel{vt} + \frac{1}{2}at^2 = \frac{1}{2}at_1^2 \rightarrow (1)$$

$$2(b) \quad x_2 = \cancel{vt} - \frac{1}{2}at^2 = -\frac{1}{2}(-b)t_2^2 \rightarrow (2)$$

$$x_2 = \frac{1}{2}b(t_2)^2$$

1 $\frac{4\alpha\beta}{(\alpha+\beta)} t^2$

2 $\frac{2\alpha\beta}{(\alpha+\beta)} t^2$

3 $\frac{\alpha\beta}{2(\alpha+\beta)} t^2$

4 $\frac{\alpha\beta}{4(\alpha+\beta)} t^2$

$$(3) \leftarrow \frac{t_1}{t_2} = \frac{b}{a}, t_1 + t_2 = t \rightarrow (4)$$

on solving (3) and (4)

$$t_1 = \left(\frac{b}{a+b}\right)t, t_2 = \left(\frac{a}{a+b}\right)t$$

$$x_1 = \frac{1}{2} a t_1^2$$

$$x_2 = \frac{1}{2} b t_2^2$$

$$x_1 + x_2 = \frac{t^2}{2} \left[a \left(\frac{b}{a+b} \right)^2 + b \left(\frac{a}{a+b} \right)^2 \right]$$

$$= \frac{t^2}{2} \left[\frac{ab^2 + ba^2}{(a+b)^2} \right]$$

$$x_1 + x_2 = \frac{t^2}{2} \left[\frac{ab(a+b)}{(a+b)^2} \right] = \boxed{\frac{abt^2}{2(a+b)}}$$



$$L = 10\text{m} \Rightarrow \frac{L_1}{L_2} = \frac{2}{3}$$

Find L_1 and L_2 .

$$\text{Sol } L_1 = \left[\frac{2}{2+3} \right] \times 10 = 4\text{cm}$$

$$L_2 = \left[\frac{3}{2+3} \right] \times 10 = 6\text{cm}$$

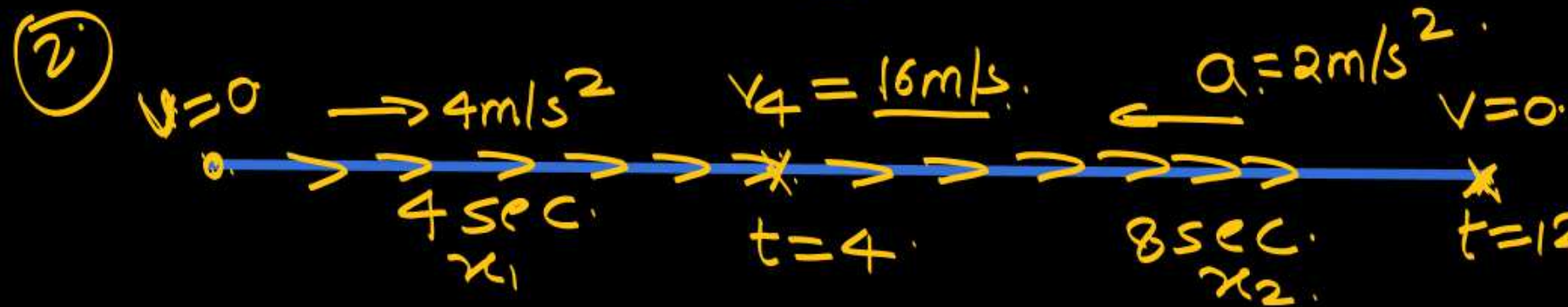
ImpX:

$$\frac{x_1}{x_2} = \frac{\frac{1}{2} a t_1^2}{\frac{1}{2} b t_2^2} = \frac{a}{b} \frac{t_1^2}{t_2^2} = \frac{t_1}{t_2} \times \frac{t_1}{t_2}$$

$$\boxed{\frac{x_1}{x_2} = \frac{t_1}{t_2}}$$

$$\frac{4}{a} = \frac{8}{4} = 2$$

A particle starting from rest has a constant acceleration of 4 ms^{-2} for 4 s. It then retards uniformly for next 8s and comes to rest.



Average speed of the particle during the motion is: $\Rightarrow t=0$ to $t=12$

1 16 ms^{-1}

2 8 ms^{-1}

3 24 ms^{-1}

4 None of these

2② $x_1 = \left[\frac{0+16}{2} \right] \times 4 = 32$, $x_2 = \left[\frac{16+0}{2} \right] \times 8 = 64$

$S = x_1 + x_2 = 32 + 64 = 96$

$\langle v \rangle = \frac{96}{12} = 8 \text{ m/s}$

$v_8 = 16 - 2 \times 4 = 8$

Magnitude of average velocity of the particle for a time interval from $t = 0$ to $t = 8 \text{ s}$ is:

1 10 ms^{-1}

2 8 ms^{-1}

3 12 ms^{-1}

4 None of these

$$\textcircled{3} \quad \langle \vec{v} \rangle = \left[\frac{v_0 + v_8}{2} \right] \times 8$$

$$\langle \vec{v} \rangle = \frac{\vec{s}}{t} = \frac{32 + 48}{8} = 10 \text{ m/s}$$

$$s_{0-4} = 32$$

$$\begin{aligned} 2 \textcircled{a} \quad s_{4-8} &= 16 \times 4 + \frac{1}{2} (-2) \times (4)^2 \\ &= 64 - 16 = 48 \end{aligned}$$

Question



$$\langle \vec{a} \rangle = \frac{\Delta \vec{v}}{t} = \frac{18 - 10}{8} = 1 \text{ m/s}^2$$

$$\langle \vec{a} \rangle = \frac{8}{8} = 1 \text{ m/s}^2$$

Magnitude of average acceleration of the particle for a time interval from $t = 0$ to $t = 8$ s is:

☒ 1 ms^{-2}

☐ 2 ms^{-2}

☐ 3 ms^{-2}

☐ 4 None of these

$$\frac{x_1}{x_2} = \frac{t_1}{t_2} = \frac{\beta}{\alpha}$$

A train accelerates from rest at a constant rate α for distance x_1 and time t_1 . After that retards at constant rate β for distance x_2 in time t_2 and comes to the rest, which of the following relation is correct:

1 $\frac{x_1}{x_2} = \frac{\alpha}{\beta} = \frac{t_1}{t_2}$

2 $\frac{x_1}{x_2} = \frac{\beta}{\alpha} = \frac{t_1}{t_2}$

3 $\frac{x_1}{x_2} = \frac{\alpha}{\beta} = \frac{t_2}{t_1}$

4 $\frac{x_1}{x_2} = \frac{\beta}{\alpha} = \frac{t_2}{t_1}$

(A) $\longrightarrow a_1$

(B) $\longrightarrow a_2$

$$s_n = \frac{a}{2}(2n-1)$$

$$s_A = \frac{a_1}{2}(2 \times 5 - 1) = \frac{9a_1}{2}$$

$$s_B = \frac{a_2}{2}(2 \times 3 - 1) = \frac{5a_2}{2}$$

$$\frac{9a_1}{2} = \frac{5a_2}{2} \Rightarrow \frac{a_1}{a_2} = \frac{5}{9} \text{ A//}$$

A body A starts from rest with an acceleration a_1 . After two seconds, another body B starts from rest with an acceleration a_2 . If they travel equal distances in the 5th second after the start of A, then the ratio of $a_1 : a_2$ is equal to

☒ 1 $5 : 9$

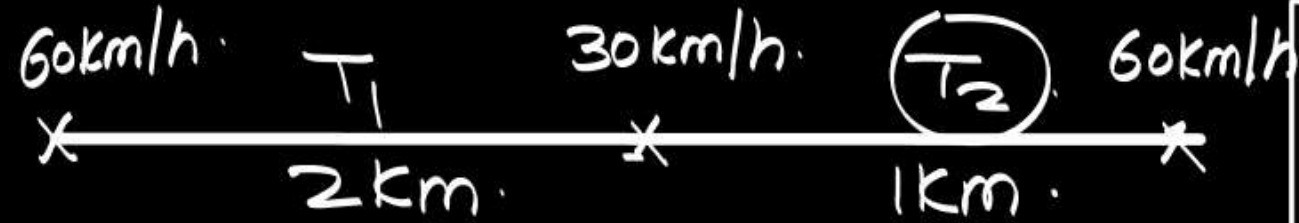
☐ 2 $1 : 3$

☐ 3 $3 : 1$

☐ 4 $9 : 1$

Question

7



60 km/h $\rightarrow$ 30 km/h $\rightarrow$ 60 km/h

$$T = \frac{3}{60} = \frac{1}{20} \text{ h} = \frac{1}{20} \times 60 = 3 \text{ min}$$

A train moving with a speed of 60 km/hr is slowed down uniformly to 30 km/hr for repair purposes during running. After this it was accelerated uniformly to reach to its original speed. If the distance covered during constant retardation be 2 km and that covered during constant acceleration be 1 km, find the time lost in the above journey

$$2 = \left[\frac{60+30}{2} \right] T_1 \Rightarrow T_1 = \frac{4}{90}$$

$$1 = \left[\frac{30+60}{2} \right] T_2 \Rightarrow T_2 = \frac{2}{90}$$

$$T_1 + T_2 = \frac{6}{90} \text{ h} = \frac{6}{90} \times 60 = 4 \text{ min}$$

1 1 min

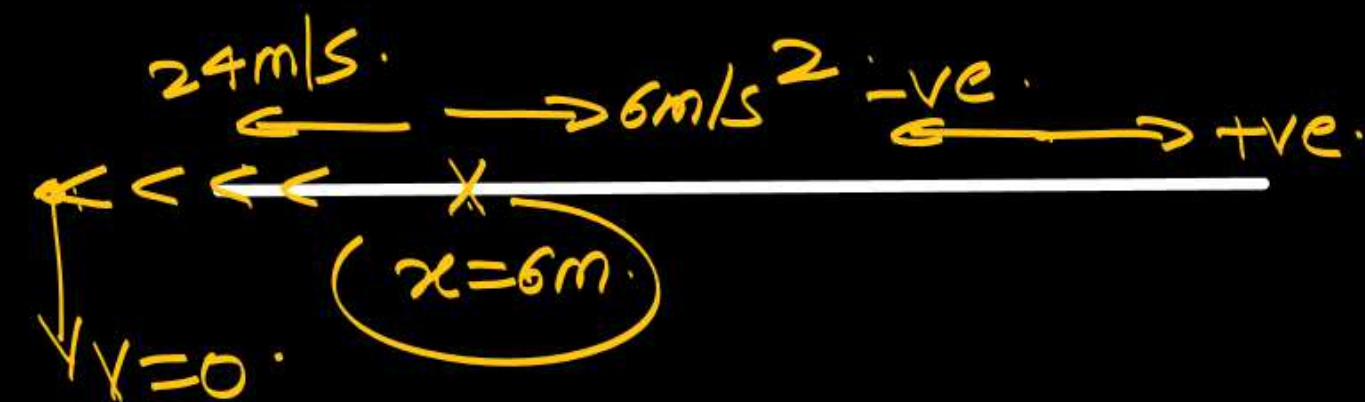
2 2 min

3 4 min

4 5 min

Question

8



An object moves along the x -axis with a constant acceleration of 6 m/s^2 and an initial velocity of -24 m/s . It is located at $x = 6 \text{ m}$ when $t = 0 \text{ s}$. What is its position when its velocity is zero?

$$\textcircled{1} \quad v = u + at$$

$$0 = -24 + 6t$$

$$t = 4 \text{ sec}$$

$$s = -24 \times 4 + \frac{1}{2} \times 6 (4)^2$$

$$x_{F-6} = s = -96 + 48 = -48, \quad x_F = -48 + 6 = -42 \text{ m}$$

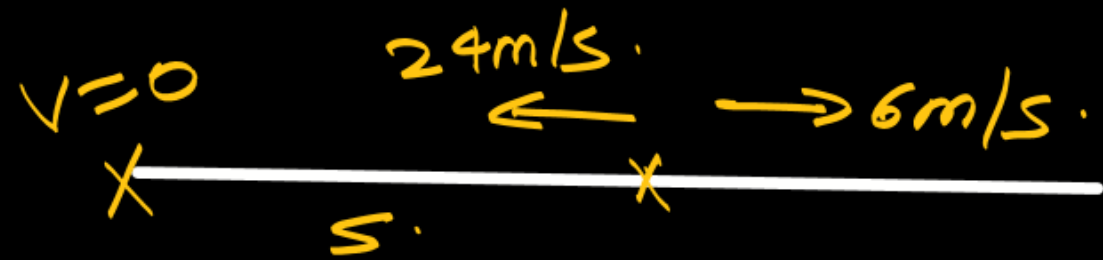
1 6 m

~~**2** -42 m~~

5 -48 m

3 -90 m

4 -96 m



$$v^2 - u^2 = 2as$$

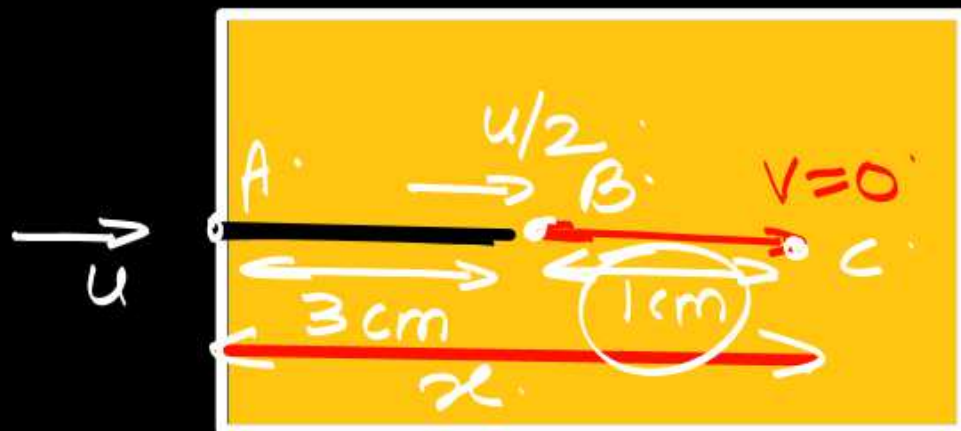
$$0^2 - (-24)^2 = 2 \times 6 \times s$$

$$\frac{-24 \times 24}{12} = s$$

$$\boxed{-48 = s}$$

→ Important (JEE MAINS)

A bullet loses half of its velocity after penetrating 3 cm into a uniform wooden block. Assuming uniform resistance, how much further will it penetrate before coming to rest?



3 → Equation of motion.

A to B

$$\left(\frac{u}{2}\right)^2 - u^2 = 2a(3) \rightarrow (1)$$

A to C

$$(0)^2 - (u)^2 = 2ax \rightarrow (2)$$

1 3 cm

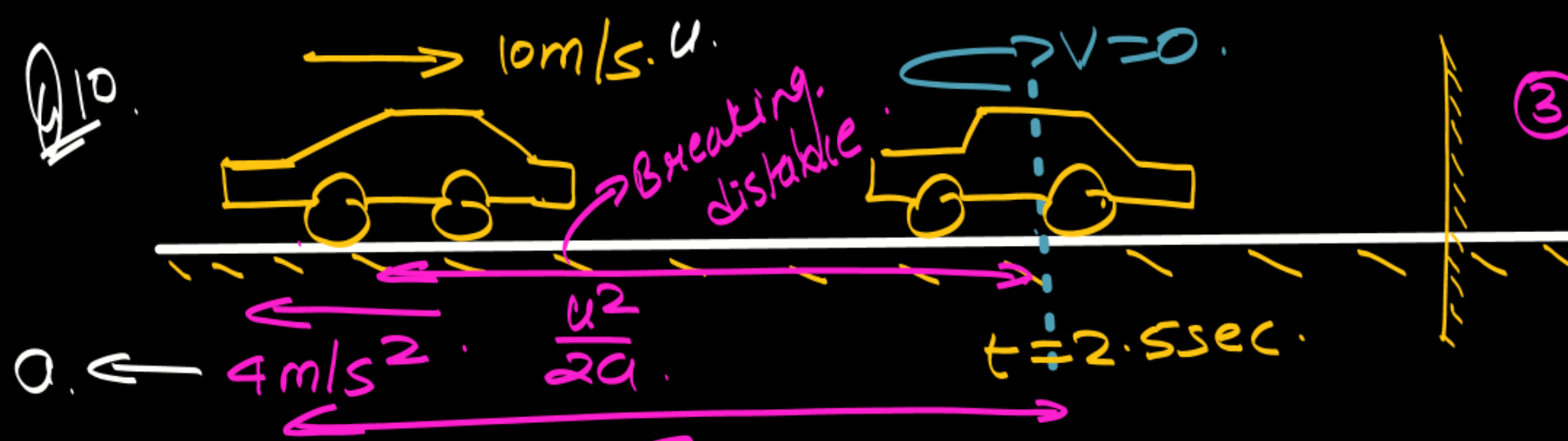
2 2 cm

3 1 cm

4 1.5 cm

$$(2) \div (1)$$

$$\frac{x}{3} = \frac{+u^2}{\frac{3u^2}{4}} = \frac{4}{3} \Rightarrow x = 4 \text{ cm}$$



③ $v^2 - u^2 = 2as$ ✓ a.
 $0^2 - (10)^2 = 2(-4)s$

$s = \frac{100}{8} = 12.5 \text{ m}$ ✓✓

$s = \frac{u^2}{2a}$ → Breaking distance.

Find distance travelled by car in 6 sec.

~~$$s = ut + \frac{1}{2}at^2$$

$$= 10 \times 6 + \frac{1}{2} \times (-4) \times (6 \times 6)$$

$$= 60 - 72 = -12 \text{ m}$$~~

→ $s = 10 \times \frac{5}{2} + \frac{1}{2} \times (-4) \times \frac{25}{4}$
 $= \frac{50}{2} - \frac{50}{4} = \frac{50}{4} = 12.5 \text{ m}$

Kitni Der mai rukh jayegii

$v = u + at$

$0 = 10 - 4 \times t \Rightarrow t = \frac{5}{2} = 2.5 \text{ sec}$

$t = 0$ to $t = 2.5$, 2.5s → 6 sec.

chal rahi thi

rukhi gayi hai

Question

Q. 11



Solution.

$\frac{u}{a}$

$$S_B = \frac{u^2}{2a}$$

$$S_B \propto u^2$$

$$u \rightarrow 2u$$

$$S \rightarrow 4S$$

$$2m \rightarrow 4 \times 2 = 8m$$

A car moving with a speed of 40 km/h can be stopped by applying brakes after at least 2 m. If the same car is moving with a speed of 80 km/h, what is the minimum stopping distance?

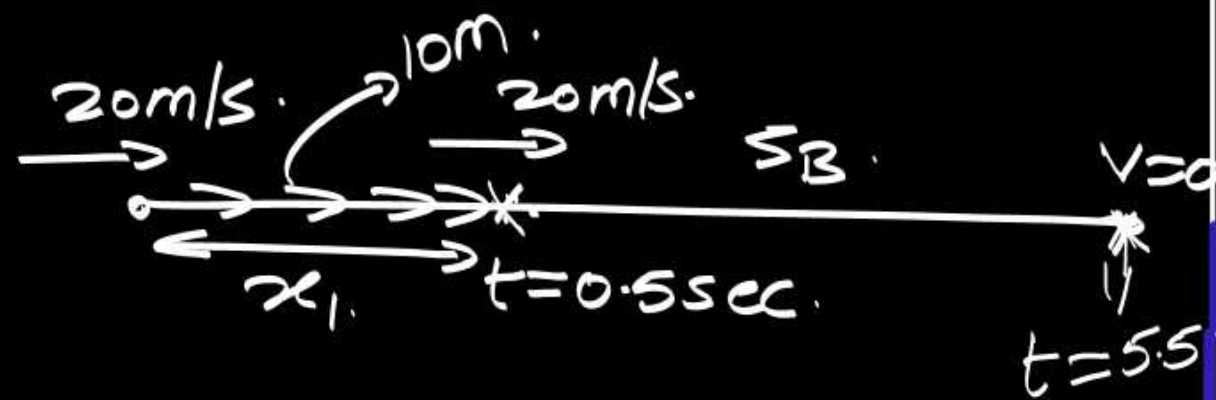
- (A) 4m ☒ (B) 8m (C) 16m (D) 32m

Question

Q. 12



$$\frac{72}{3.6} \times \frac{5}{18} = 20 \text{ m/s}$$



$$x_1 = 20 \times \frac{1}{2} = 10 \text{ m}$$

↓ uniform motion ($a=0$)

$$S_B = \frac{u^2}{2a} = \frac{10^2}{2 \times 4} = 50 \text{ m}$$

$$S = 10 + 50 = 60 \text{ m}$$

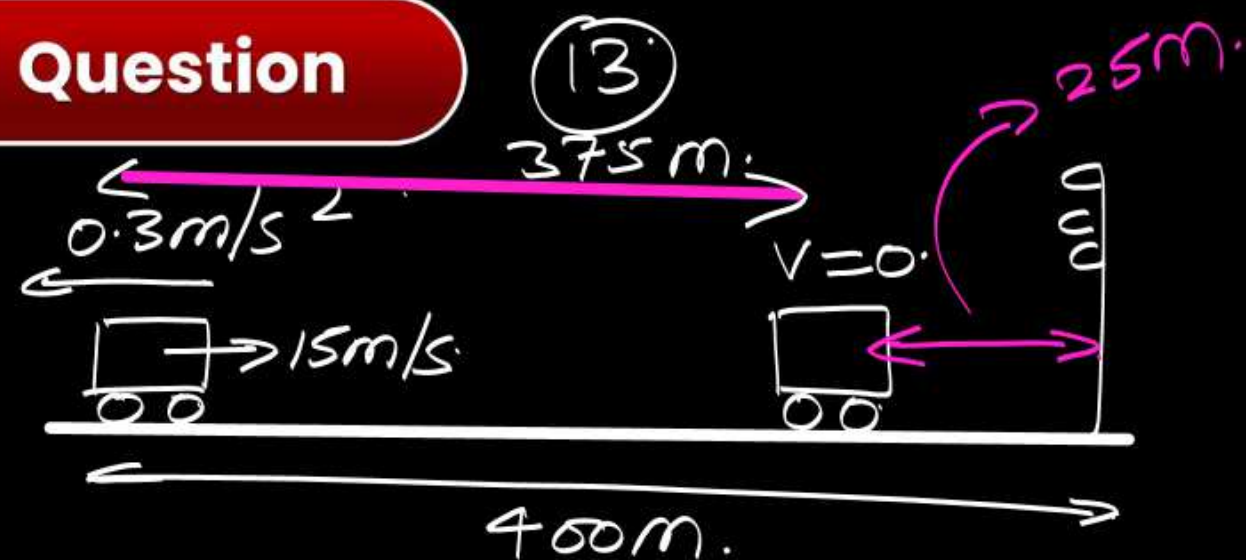
A car is moving with speed 72 kmph. At $t = 0$ driver see that traffic light turn red and apply breaks thereafter. Reaction time of driver is 0.5 sec and break produce constant retardation of 4 m/s^2 . Find time to stop and final distance of car travelled.

$$0 = 20 - 4 \times t$$

$$t = \frac{20}{4} = 5 \text{ sec}$$

$$t = 5.5 \text{ sec}$$

Question



$$0 = 15 - (0.3) \times t$$

$$t = \frac{15}{0.3} = \frac{150}{3} = 50 \text{ s}$$

1st

$$s_B = \frac{u^2}{2a} = \frac{15 \times 150}{2 \times 0.3} = \frac{5 \times 75}{\cancel{15} \times \cancel{10}} = 375 \text{ m}$$

$$s = 400 - 375 = 25 \text{ m}$$

At a distance $L = 400 \text{ m}$ from the traffic light brakes are applied to a locomotive moving at a velocity $v = 54 \text{ km/hr}$. Determine the position of the locomotive relative to the traffic light 1 minute after the application of the brakes if its acceleration is -0.3 m/sec^2 .

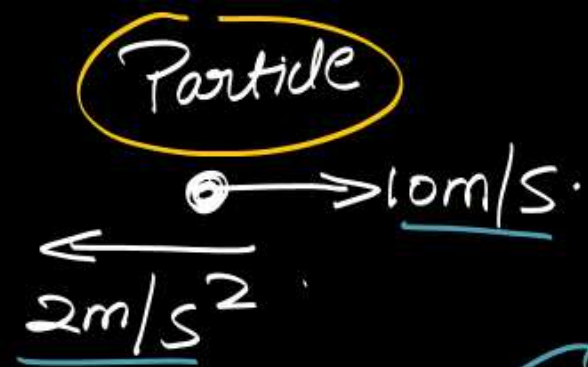
$$\frac{3}{54} \times \frac{5}{18} = 15 \text{ m/s}$$

$$s = ut + \frac{1}{2}at^2$$

$$s = 15 \times 50 + \frac{1}{2} \times \left[\frac{-3}{10} \right] \times 50 \times 50$$

$$s = 750 - (125 \times 3) = 375$$

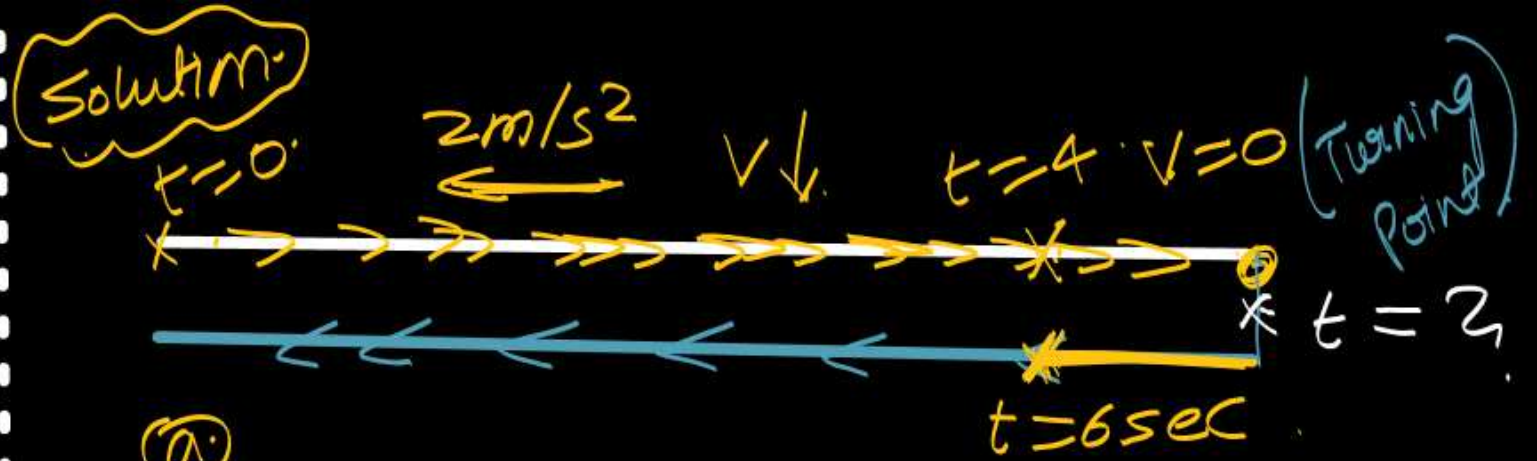
(14)



Find.

- (a) time of turning point.
- (b) Distance and Displacement in 4 sec.
- (c) Distance and Displacement in 6 sec.

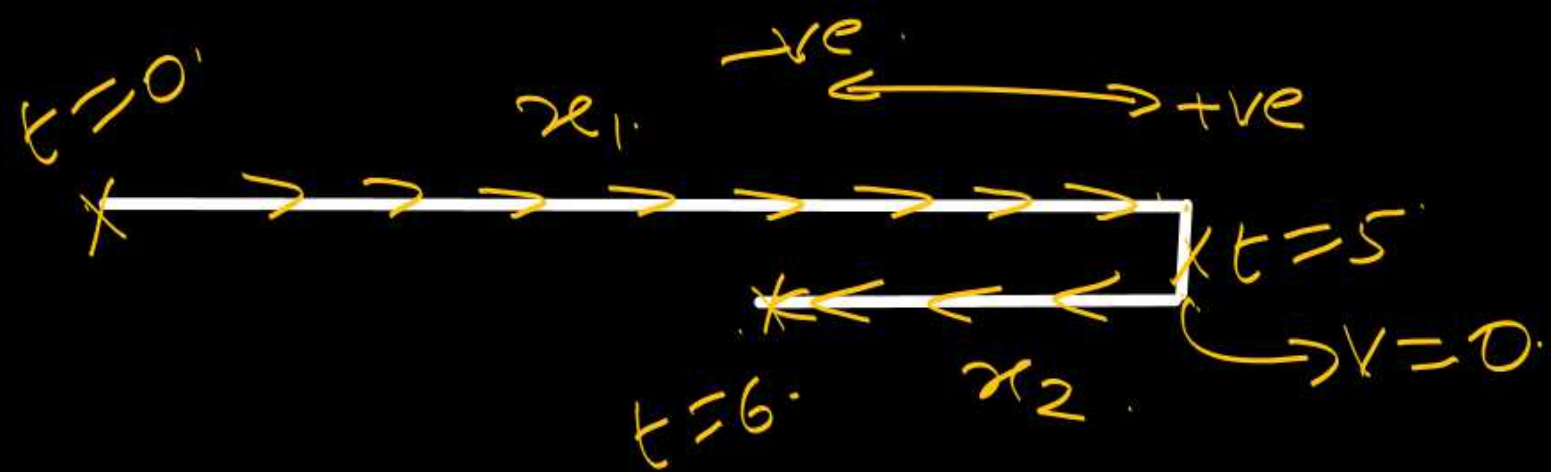
$t = 5 \text{ sec}$



Solⁿ (a) $v = u + at$
 $0 = 10 - 2 \times t \Rightarrow t = 5 \text{ sec}$

(b) $\vec{s} = ut + \frac{1}{2}at^2$
↓
Displacement
 $\vec{s}_{0-4} = 10 \times 4 + \frac{1}{2}(-2) \times (4)^2 = 24 \text{ m}$
 $s_{0-4} = 24 \text{ m}$

(c) $\vec{s}_{0-6} = 10 \times 6 + \frac{1}{2}(-2) \left(\frac{36}{6}\right) = 24 \text{ m}$



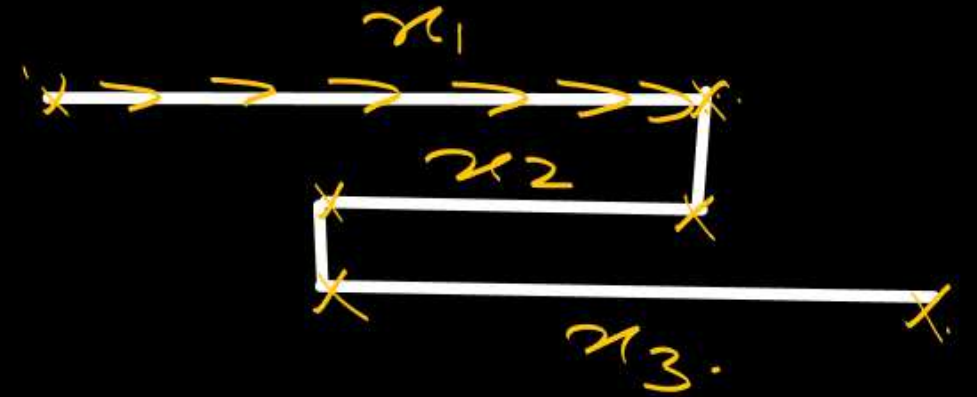
$$x_1 = x_{0-5} = 10 \times 5 + \frac{1}{2} (-2) \times (25)$$

$$= 50 - 25 = 25 \text{ m.}$$

$$x_{5-6} = 0 \times 1 - \frac{1}{2} \times (-2) (1)^2$$

$$x_2 \quad x_{5-6} = -1 \text{ m.}$$

$$S = |x_1| + |x_2| = 25 + 1 = \underline{\underline{26 \text{ m}}}$$



$$S = |x_1| + |x_2| + |x_3|$$

A particle has initial velocity 10 m/s is moving in straight line, due to constant retarding force along the line of velocity which produces retardation of 5 m/s^2 ,

- 1** The maximum displacement in the direction of initial velocity is 10 m .
- 2** The distance travelled in first 3 seconds is 7.5 m .
- 3** The distance travelled in first 3 seconds is 12.5 m .
- 4** The distance travelled in first 5 seconds is 17.5 m .

For a particle moving along a straight line, mark the CORRECT statements

- 1** Distance travelled by particle is always equal to magnitude of displacement.
- 2** Average speed is always equal to magnitude of average velocity measured for a particular time interval.
- 3** Instantaneous speed is always equal to magnitude of instantaneous velocity.
- 4** If velocity of a particle is zero then acceleration of particle must be zero at that instant.

Which of the following statement(s) is/are correct?

- 1** A body can have constant speed but varying velocity.
- 2** A body can have constant velocity but varying speed.
- 3** A body can have non-zero acceleration with constant velocity.
- 4** A body can have velocity without having acceleration.

A runner maintains a constant speed of 6 m/s around a perfectly circular track. Which of the following is true regarding her velocity?

- 1** The velocity is constant because the speed is constant.
- 2** The velocity is continuously changing because the direction of motion is changing.
- 3** The average velocity for any partial lap will always be zero.
- 4** The acceleration of the runner is zero since the speed does not change.

A train, travelling at 20 km/hr is approaching a platform. A bird is sitting on a pole on the platform. When the train is at a distance of 2 km from pole, brakes are applied which produce a uniform deceleration in it to stop it at pole. At that instant the bird flies towards the train at 60 km/hr and after touching the nearest point on the train flies back to the pole and then flies towards the train and continues repeating itself. Calculate how much distance will the bird have flown before the train stops?

Paragraph For Next Two Questions

For any particle moving with some velocity ($\vec{v}$) and acceleration ($\vec{a}$), tangential acceleration and acceleration are defined as follows

Tangential acceleration- The component of acceleration in the direction of velocity.

Normal acceleration- The component of acceleration in the direction perpendicular to velocity.

If at a given instant, velocity and acceleration of a particle are given by.

$$\vec{v} = 4\hat{i} + 3\hat{j}$$

$$\vec{a} = 10\hat{i} + 15\hat{j} + 20\vec{k}$$

Find the tangential acceleration of the particle at the given instant:

1 $17(4\hat{i} + 3\hat{j})$

2 $\frac{17}{5}(4\hat{i} + 3\hat{j})$

3 $17(4\hat{i} - 3\hat{j})$

4 $\frac{17}{5}(4\hat{i} - 3\hat{j})$

Find the normal acceleration of the particle at the given instant:

1 $\frac{-9\hat{i}+12\hat{j}+50\hat{k}}{5}$

2 $\frac{9\hat{i}-12\hat{j}-50\hat{k}}{5}$

3 $\frac{-18\hat{i}+24\hat{j}+100\hat{k}}{5}$

4 $\frac{18\hat{i}+-24\hat{j}-100\hat{k}}{5}$



Homework



① Revise class notes.

② Attempt all questions of PPT. (7-questions).

③ DPP → 05.

④ Module → Chapter → 02 → Page No 27 → Examples (1 to 7).
Ex-02 (Prabal) - [16, 22, 25]

Thank

You